

A 28-nm RRAM Computing-in-Memory Macro Using Weighted Hybrid 2T1R Cell Array and Reference Subtracting Sense Amplifier for AI Edge Inference

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Abstract—Non-volatile computing-in-memory (nvCIM) can potentially meet the ever-increasing demands on improving the energy efficiency (EF) for intelligent edge devices. However, it still suffers from limited input parallelism due to the parasitic effects, signal margin degradation due to device non-idealities, and large hardware cost for analog readout. In this work, we present a two-transistor-one-resistor (2T1R) resistive memory (RRAM) nvCIM macro featuring: 1) a macro structure with decoupled memory and computing data paths; 2) the weighted hybrid 2T1R (WH-2T1R) cell array; 3) the redundant sub-array mapping scheme of the most-significant-bit (RSM-MSB); and 4) reference-subtracting current sense amplifier (RS-CSA). A test-chip is silicon-verified using the 28-nm high-k/metal-gate (HKMG) logic process with foundry-developed RRAM. The test-chip performs linear analog multiply-and-accumulate (MAC) operations over 32 accumulation channels and achieves 30.34–154.04 TOPS/W with 1-bit input (IN), 3-bit weight (W), and 4-bit output (O). Evaluations with the ResNet-18 model show that the MSB-RSM scheme results in 0.96% and 2.83% improvement on CIFAR-10 and CIFAR-100 inference accuracy, respectively.

Index Terms—Analog readout, artificial intelligence (AI), computing-in-memory, edge computing, resistive memory (RRAM), sense amplifier (SA).

I. INTRODUCTION

THE rapid development of the Internet of Things (IoT) calls for the relentless improvements on computing energy efficiency (EF) to sustain the exponentially increasing demands on edge intelligent devices [1], [2], [3], [4], [5], [6]. Non-volatile computing-in-memory (nvCIM) offers a competent solution to develop artificial intelligence (AI) edge processors with a high EF by accommodating the weight data on-chip and performing multiply-and-accumulate (MAC) operations directly inside the memory arrays [7], [8], [9], [10], [11], which minimizes the power and latency due to the data movement between the computing and memory units as well as that between the hierarchical memories. Note that the non-volatile property is particularly attractive for energy-harvesting (EH) intelligent monitoring and sensing devices, as it substantially reduces the power for weight data back-up and recovery, and hence improves their real-time performance, as shown in Fig. 1.

Resistive memory (RRAM) nvCIMs are being intensively investigated because of RRAM's high density, low power, compatibility with advanced CMOS, and good cost-effectiveness [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. Previous arts have already paid considerable research endeavors to develop fully integrated RRAM nvCIM chips based on the production-ready foundry RRAMs [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39]. However, several remaining issues still exist, as shown in Fig. 2(a). First, because of the increased non-linearity of the MAC signals as the input parallelism increases, most of these chips suffer from a limited number (≤ 16) of analog accumulation channels [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39]. Second, the small MAC signal margins limit the computing precision in the analog domain, and hence many nvCIM chips rely on the digital shift-and-add circuits for multi-bit weight processing

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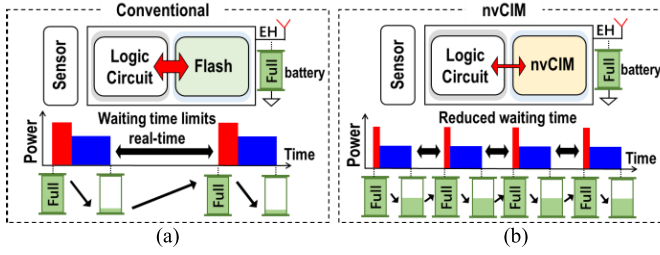


Fig. 1. Typical EH edge intelligent devices and their real-time performance. (a) Conventional NVM case. (b) nvCIM case.

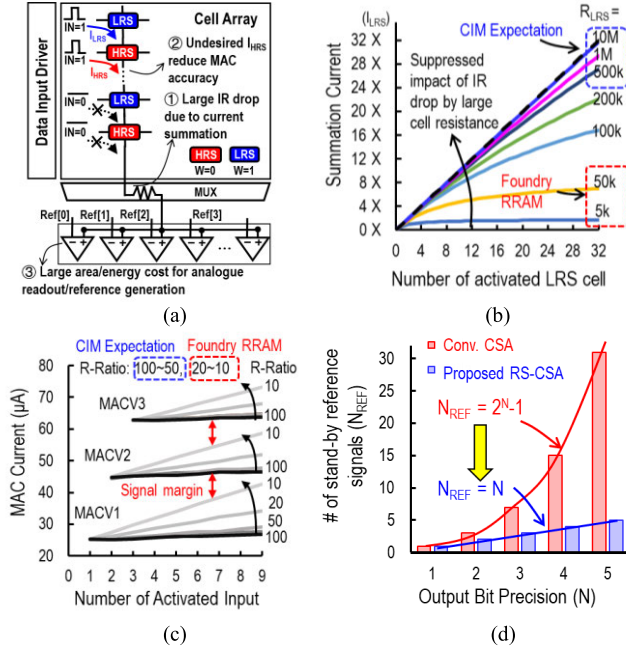


Fig. 2. Major challenges in RRAM nvCIM designs. (a) Conceptual view of the challenges. (b) MAC signal non-linearity with different cell resistances. (c) MAC signal margin degradation with different R -ratios. (d) Number of stand-by reference signals for different output precision.

[32], [33], [36], [37]. Finally, the large energy cost and area cost of the analog readout circuits considerably degrade the macro performance. Particularly, a large number of stand-by reference signals are involved in several designs, yet the reference generation is less discussed [40], [41], [42], [43]. To address these challenges, we present a two-transistor-one-resistor (2T1R) RRAM nvCIM macro verified using the 28-nm high- k /metal-gate (HKMG) logic process with foundry-developed RRAM.

The reminder of this work is organized as follows. Section II further discusses the root causes of the nvCIM design challenges and presents the proposed macro. Section III describes the proposed weight hybrid 2T1R cell array (WH-2T1R) and its weight mapping strategy for highly efficient analog MAC operations. Section IV presents the proposed reference-subtracting current sense amplifier (RS-CSA) for analog readout with reduced reference generation costs and improved dynamic linear range. Section V shows the measurement and evaluation results. Section VI concludes this article.

II. DESIGN CHALLENGES AND THE PROPOSED RRAM nvCIM

A. Large MAC Non-Linearity for High Input Parallelism

Fig. 2(b) shows the increased non-linearity of the summed low-resistive-state (LRS) cell current (I_{LRS}) as the number of activated LRS cell increases (N_{LRS}) in a typical current-type nvCIM. It can be attributed to the IR drops due to the parasitic resistances on the data path. Note that although the recently proposed nvCIMs that perform computations in voltage [33], [41], [43], charge [44], or time domain [37] can effectively reduce the summation current, the non-linearity effect is still a major concern in these works as involving parasitic resistance tends to change the voltage division or the charge transferring behaviors.

To suppress the non-linearity effect, a large cell resistance at the M Ω level is expected in nvCIM to suppress the impacts caused by the parasitic resistance [10]. However, increasing cell resistance is non-trivial because the typical filamentary-type RRAM cells based on transition metal oxides developed in major foundries have a fixed LRS resistance ranging from 5 to 50 k Ω [16], [17], [18], [19], [20], [21], [22]. Besides, nvCIM needs to be operated in both the memory and CIM modes. A large cell resistance tends to degrade the readout latency in the memory mode, though it is favored in the CIM mode.

B. Small MAC Signal Margin for Multi-Bit Processing

The premise of performing multi-bit MAC operations is ensuring sufficient signal margins. One of the primary concerns is the margin degradation due to data-pattern variations. Fig. 2(c) shows the distributions of different MAC values (MACVs) versus the input data patterns as a function of resistance ratio (R -ratio). The MACV margins degrade severely as the number of activated inputs increases for small R -ratio devices. This is because although the product value (PV) of weight-0, represented by the high-resistive-state (HRS) cell, and input-1, represented by the activated input, should be 0 theoretically, there is a non-negligible HRS cell current (I_{HRS}) in practical. Besides, process variations and non-uniformly distributed resistance levels of the multi-level cells (MLC) [38], [45] further deteriorate the MAC signal margin particularly for multi-bit MAC operations.

Although a large cell R -ratio can suppress the non-idealities and improve the signal margins, it becomes problematic considering the intrinsic tradeoff between R -ratio and endurance. Most foundry RRAM cells are developed as single-level cells (SLCs) operated with R -ratios ranging from 5 to 20 for sufficient endurance [12], [16], [33], which hardly meet the CIM requirements. Previous works have proposed using two-transistor-two-resistor (2T2R) differential cell [26], input-dependent sensing scheme [27], [33], and HRS compensation circuit to suppress the impacts of HRS leakage [31], yet tend to suffer from considerable area or latency overhead. Besides, to process multi-bit MAC in the analog domain, previous works have proposed to use ratioed current mirrors (CMs) [28] or different clamping voltages to represent multi-bit weights [29], which tends to incur additional energy cost.

C. High Energy and Area Costs for Analog Readout

The widespread and narrow-spaced MAC signals in the nvCIM macro have posed a great challenge for the analog readout circuitry. High-precision analog-to-digital converters (ADCs) have large energy and area costs, which not only limit the computing EF but also degrade the macro density and the output parallelism. In particular, the reference generator in ADCs can account for a large part of total energy and area. As shown in Fig. 2(d), $2^N - 1$ reference signals are required for N -bit readout in the conventional analog readout circuitry.

To address this issue, low-power and small-area multi-bit sense amplifiers (SAs) with small offsets, enhanced margins, and widened input ranges are expected to enable high EF nvCIM macros. Previous works have proposed low-power SAs with offset cancellation and margin enhancement techniques [27], [28], [29], [40]. However, the techniques to increase the SA input range are less discussed. As for reference generations, previous works have proposed to use dedicated dummy cell arrays [27], resistive dividers based on poly-silicon resistors [35], or capacitive digital-to-analog converters (DACs) [26] for reference generation, which are area- and power-consuming.

D. Proposed WH-2T1R RRAM nvCIM Macro

To overcome the above challenges, we proposed a 2T1R RRAM nvCIM macro featuring: 1) a macro structure with decoupled memory and computing data paths to reduce the parasitic effects on the CIM data path; 2) the weighted hybrid 2T1R (WH-2T1R) cell array to optimize the input parallels and signal margin for multi-bit MAC operations; 3) the redundant sub-array mapping scheme of the most-significant-bit (MSB-RSM) to suppress devices variations; and 4) RS-CSA to extend the SA input range and reduce the costs for reference generation.

As shown in Fig. 3(a), the proposed macro consists of a 64×128 WH-2T1R cell array, addressing circuits for word-line (WL), bitline (BL), source-line (SL), and transpose BL (TBL), decoupled data paths, and input-output (I/O) circuits for the memory and CIM modes. Its overall operations are illustrated in Fig. 3(b). In the memory mode (CIMB = 1), the macro can be operated as the conventional 1T1R array. The RRAM cells can be addressed by the row (AX) and column address (AY) to write/read (WEB = 0/1) the I/O data (DIN/DOUT). In the CIM mode (CIMB = 0), it performs analog MAC operations with 1-bit input (IN), 3-bit weight (W), and 4-bit output (O) over 32 accumulation channels in a given clock (CLK) cycle. The CIM output data (COUT) can be readout from the TBL current (I_{TBL}) by the RS-CSA. Using decoupled data paths in the memory and CIM modes, the macro can fulfill their different requirements on the cell currents and R -ratios. Besides, while the memory data path and periphery need I/O devices to tolerant the RRAM write voltage, the CIM ones can be built by logic core devices to reduce the parasitic effects. Compared with the conventional 1T1R counterpart, the proposed design can comprehen-

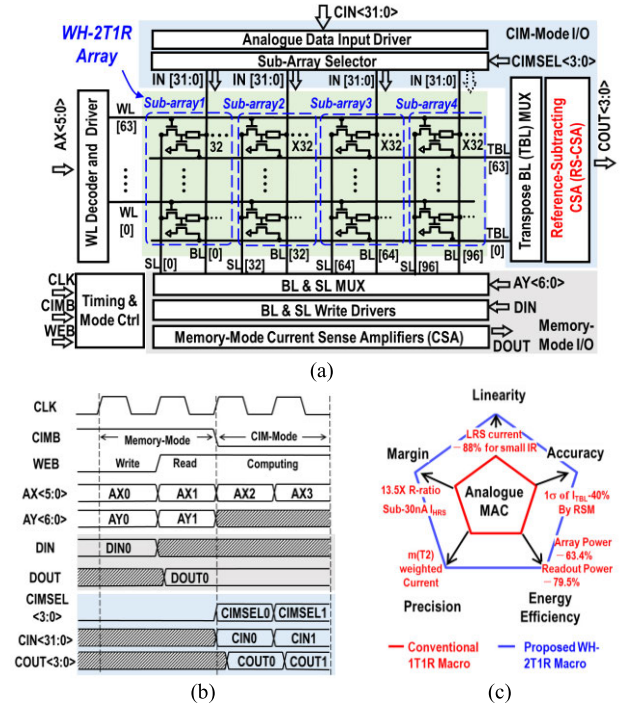


Fig. 3. Proposed 2T1R RRAM nvCIM macro. (a) Macro structure. (b) Operational waveforms. (c) Summary of major advantages.

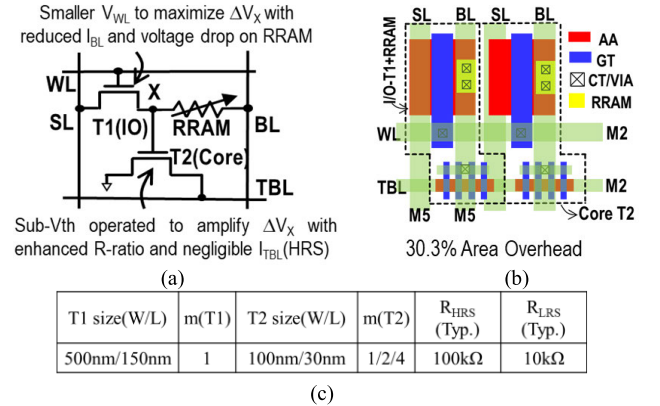


Fig. 4. (a) WH-2T1R bit-cell structure. (b) Bit-cell layout with $m(T2) = 4$. (c) Key parameters of the bit-cell.

sively improve the performance of analog MAC operations [Fig. 3(c)], which are discussed in detail in Section III.

III. WEIGHTED HYBRID 2T1R CELL ARRAY

A. Weighted Hybrid 2T1R (WH-2T1R) Bit-Cell

The WH-2T1R bit-cell structure is shown in Fig. 4(a). It hybridizes an IO device (T1) as the memory selector and a core device (T2) for the CIM read. While T1 is used to tolerate the voltages for set and reset in write operations, T2 is used to amplify the R -ratio, weight the cell current, and reduce the parasitic resistance and capacitance in the CIM data path for MAC operations. The IO T1 and core T2 are vertically aligned with an area overhead of 30.3% compared with the 1T1R bit-cell, as shown in Fig. 4(b). It is also worthy to note that to achieve a small area overhead, the size or the number of

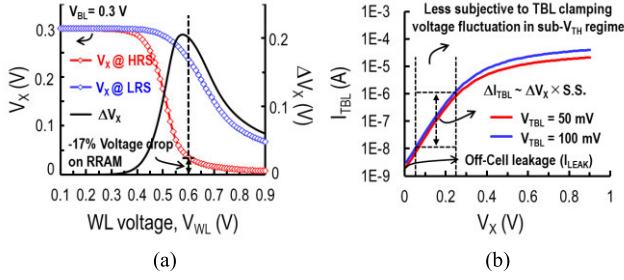


Fig. 5. Biasing scheme simulations for WH-2T1R cell. (a) V_X at HRS and LRS and ΔV_X as a function of V_{WL} . (b) I_{TBL} versus V_X with different V_{TBL} .

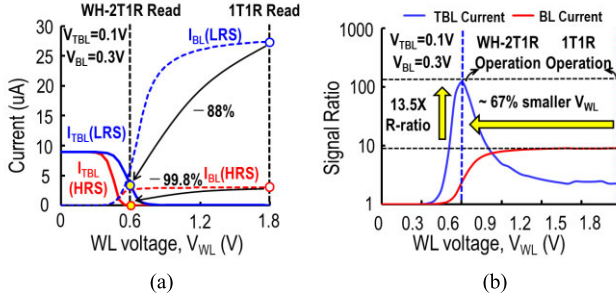


Fig. 6. Simulated read operations of the proposed WH-2T1R cell and the 1T1R cell. (a) Cell currents. (b) Signal ratios and operational voltage.

multipliers of T2 [$m(T2)$] should be limited to a certain range so that the metal line pitches in the cell array are dominated by T1. In this work, $m(T2)$ varies from 1, 2, and 4 for 3-bit weighting in the analog domain.

The signal ratio of the WH-2T1R cell can be maximized by optimizing the biasing scheme. Fig. 5(a) shows the voltage at the node X (V_X) changes with varying WL voltage (V_{WL}), which is maximized by half-activating T1 to the point that the resistance of T1 (R_{T1}) equals to the square root of the product of the cell resistance at HRS (R_{HRS}) and LRS (R_{LRS}). Note that because of the half-activated T1, the voltage drops on the RRAM cells, defined by the difference between the BL voltage (V_{BL}) and V_X , can be reduced by 17% compared with that in the case that T1 is fully activated, which can suppress the read disturb issue. On the other hand, T2 is operated at the sub-threshold region to amplify ΔV_X leveraging its steep sub-threshold swing (S.S.), as shown in Fig. 5(b). Note that I_{TBL} is less dependent on the TBL voltage (V_{TBL}) in the sub-threshold region than that in the linear region, which makes I_{TBL} less subjective to clamping voltage fluctuations. However, there is an off-cell leakage current (I_{LEAK}) in T2, which becomes non-negligible as the array size increases and needs to be taken into considerations in the readout circuitry design.

Fig. 6 shows comparison of the cell currents, R -ratios, and operational voltages of the read operations of the conventional 1T1R cell and the proposed WH-2T1R cell. The proposed scheme can reduce the LRS current by 88%, effectively suppress the HRS leakage current, enhance the R -ratio by 13.5 times, and reduce V_{WL} by 67%, which supports highly parallel and low-power MAC operations.

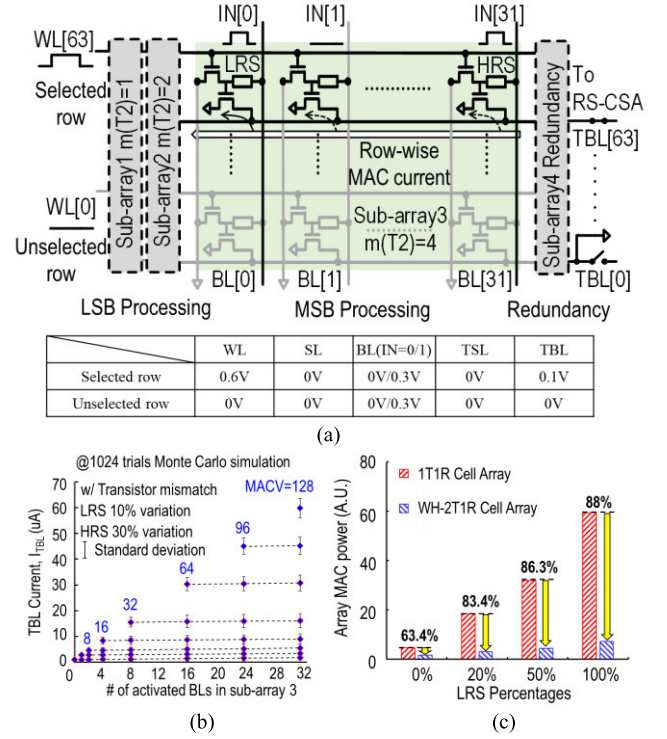


Fig. 7. (a) MAC operations with WH-2T1R cell array and its biasing conditions. (b) Simulated I_{TBL} distributions for different MACVs. (c) Simulated array MAC power.

B. MAC Operations and Redundant Weight Mapping Scheme

Fig. 7(a) shows the 64×128 WH-2T1R cell array comprising four 64×32 sub-arrays (SUBAs). Three SUBAs (SUBA1-3) are used to store 3-bit weight data for analog MAC, in which different numbers of multiplier ($m = 1$, $m = 2$, and $m = 4$) of T2 in the WH-2T1R cell are used to represent different place values of the 3-bit weight. One additional sub-array (SUBA4) with $m = 4$ is used as the redundancy for the most-significant-bit (MSB). In the CIM mode, while the inputs are columnwisely applied through BLs, the analog MACs are performed rowwisely through TBLs. The WL and TBL are activated or grounded for the selected or unselected rows, respectively. The inputs and weights' encoding scheme is summarized in Table 1. Fig. 7(b) shows the typical MAC behaviors using SUBA3. Because of the reduced cell currents and the enhanced R -ratio, the non-linearity of MAC signals and the signal margin degradations due to I_{HRS} can be well suppressed. Besides, the WH-2T1R cell array exhibits more than 63.4% reduced MAC power compared with the 1T1R array, as shown in Fig. 7(c).

To suppress the impact of device variations, we have further proposed the redundant subarray mapping (RSM) scheme, as shown in Fig. 8. To perform the scheme, SUBA4 is activated by the sub-array selector and the inputs data are sent to the regular SUBA and the redundant one. On one hand, MSB-0 is mapped to HRS cells in both regular and redundant SUBAs. On the other hand, weight-1 is first mapped to LRS cells in both SUBAs. Then we calculate the derivations (Δ_1 and Δ_2) in the two SUBAs, which are defined by the difference between the cell currents (I_{cell1} and I_{cell2}) and the target current (I_{Target}).

TABLE I
INPUT AND WEIGHT DATA ENCODING SCHEME

IN[i]	SL/BL[i] in array-1/2/3	3-bit Weight	Cell[i] in array-1	Cell[i] in array-2	Cell[i] in array-3	Product (IN × W)	TBL current (I_{TBL})
0	GND	XXX	-	-	-	000	0
1	SL=GND; BL[i]= $V_{BL}(0.3V)$	000	HRS	HRS	HRS	000	Negligible
		001	LRS	HRS	HRS	001	$I_{TBL}(LRS)$
		010	HRS	LRS	HRS	010	$2 * I_{TBL}(LRS)$
		011	LRS	LRS	HRS	011	$3 * I_{TBL}(LRS)$
		111	LRS	LRS	LRS	111	$7 * I_{TBL}(LRS)$

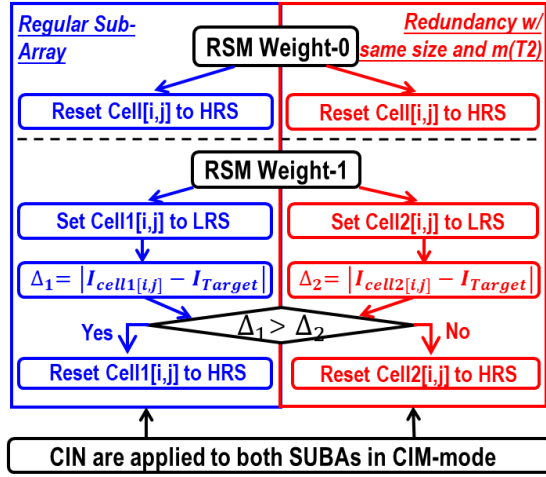


Fig. 8. Operation flow of the proposed redundant sub-array mapping (RSM) scheme.

The one with a smaller Δ is kept and the other one is reset back to the HRS. Note that the preconditions of the RSM scheme are the large R -ratio and the negligible I_{HRS} , otherwise it may incur additional errors due to the dependence of MACV signals on the input patterns.

IV. PROPOSED RS-CSA

A. Structure and Operational Flow of the Proposed RS-CSA

Fig. 9(a) shows the structure of the proposed RS-CSA. It uses a current mirror (CM) to compare the data-line current (I_{DL}) with the reference current (I_{REF}) and output COUT[0:3]. On the reference (REF) side, I_{REF} is determined by four REF branches with ratioed reference currents ($I_{REF}[0] = 2I_{REF}[1] = 4I_{REF}[2] = 8I_{REF}[3]$) controlled by SEN[0:3]. On the data-line (DL) side, the DL voltage is clamped to the clamping voltage (V_{CLP}). Besides, four additional branches are used to modulate the input (I_{IN}) and output currents (I_{OUT}) of the CM, including a leakage current (I_{LEAK}) compensation branch to compensate the off-cell leakage in the cell array controlled by SW[0] and three reference-subtracting (REFS) branches that can subtract $I_{REF}[0]$, $I_{REF}[1]$, or $I_{REF}[2]$ from the data-line current (I_{DL}) depending on SW[1:3]. The RS control determines SEN[0:3] and SW[0:3] in different operation phases, as shown in Fig. 9(b). The reference voltages for REF (V_{REF_N}) and REFS (V_{REF_P}) are generated based on

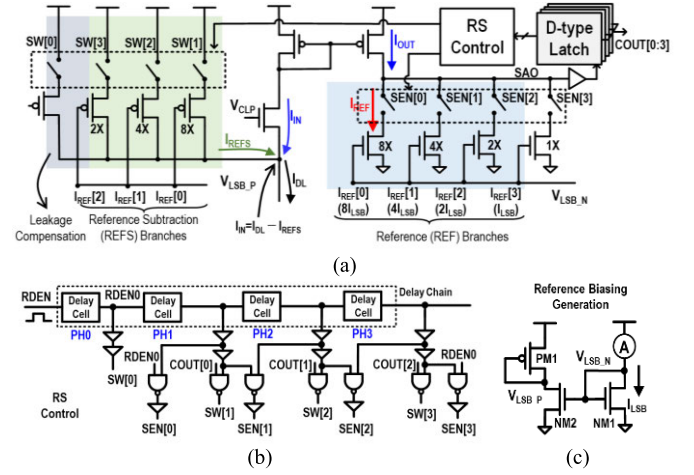


Fig. 9. Proposed RS-CSA. (a) Overall structure. (b) Reference-subtracting (RS) control. (c) Biasing generation for the REF and REFS branches.

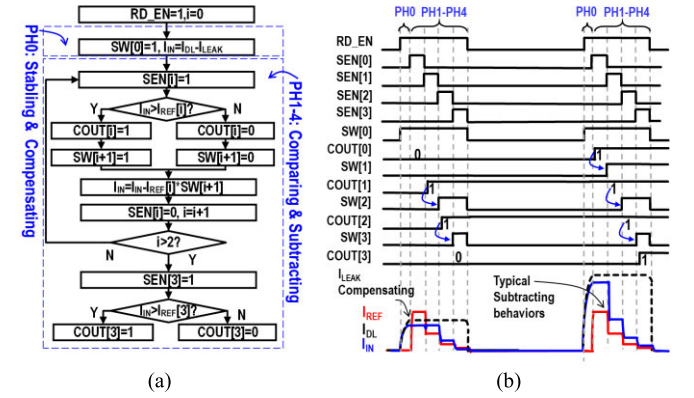


Fig. 10. Operation of RS-CSA. (a) Flowchart. (b) Operational waveforms.

the lowest-significant-bit current (I_{LSB}), as shown in Fig. 9(c). After applying I_{LSB} , the diode-connected NM1 converts I_{LSB} into V_{REF_N} . I_{LSB} is duplicated to the other branch of the current mirror and the diode-connected PM1 converts it into V_{REF_P} . Note that the gate voltage (V_{LEAK}) of the PMOS in the leakage compensation branch and I_{LSB} are controlled by the off-chip instrument in our current implementation for the test purpose.

Fig. 10 illustrates the RS-CSA operation. A typical readout process, triggered by RDEN, consists of one phase (PH0) for current stabling and leakage compensating and four phases (PH1-PH4) for current comparing and reference subtracting. In PH0, SW0 is triggered to compensate I_{LEAK} during I_{DL} stabling ($I_{COMP} = I_{DL} - I_{LEAK}$). This step is of particular importance for the large cell array built with the high-performance logic transistors, which tend to contribute a non-negligible I_{LEAK} to the computing path and result in computing errors. In PH1-4, I_{COMP} is sequentially compared with $I_{REF}[0:3]$ to output COUT[0:3], in which the RS control cumulatively subtracts $I_{REF}[i - 1]$ at the i th phase from I_{DL} if $I_{COMP} > I_{REF}[i - 1]$ till the last one. By this means, the RS-CSA is able to output the 4-bit COUT with four stand-by reference signals.

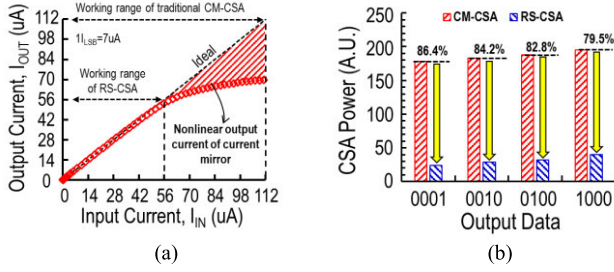


Fig. 11. Comparisons between the conventional CM-CSA and the RS-CSA. (a) Simulated CM input and output currents. (b) Simulated power.

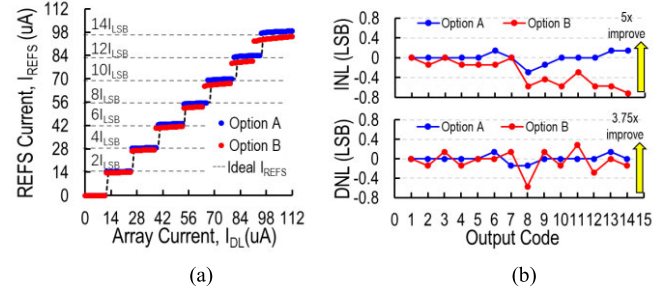


Fig. 13. Simulated comparison results of different RS configurations. (a) REFS current versus I_{DL} . (b) INL and DNL of different output codes.

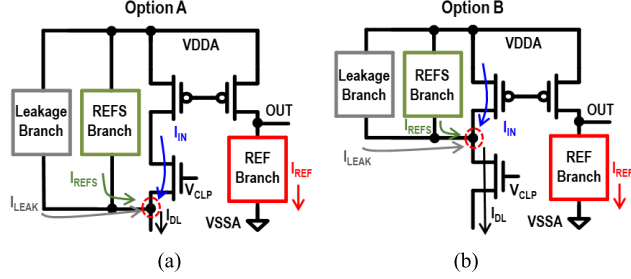


Fig. 12. Two different configurations for reference subtracting. (a) RS after voltage clamping. (b) RS before voltage clamping.

B. Reference-Subtracting Behaviors

Compared with the conventional CM-based current SA (CM-CSA) without RS, the advantages of the proposed RS-CSA are twofold. First, the RS-CSA can halve the CM input current range for the same I_{DL} range, which makes it work in the dynamic linear range of the CM to suppress the non-linearity errors [Fig. 11(a)]. Second, the 4-bit RS-CSA can reduce the number of stand-by reference signals from 16 to 4, which achieves more than 79.5% power saving across different output patterns [Fig. 11(b)]. Note that although the bit serial readout manner of the RS-CSA tends to increase the readout latency, this side effect can be bypassed by placing more RS-CSAs on chip to improve the computing parallelism because the RS-CSA can achieve reduced power and area costs compared with its conventional counterpart.

To suppress the readout errors in RS-CSA, it is key to select the RS nodes in the DL. One option (Option-A) is performing RS after voltage clamping [Fig. 12(a)], and the other one (Option-B) is carrying out RS before voltage clamping [Fig. 12(b)]. Although both of them can achieve the RS functions, they result in considerable differences in quantization accuracy. Fig. 13(a) shows the REFS current (I_{REFS}) as a function of I_{DL} . Compared with that in option-A, I_{REFS} shows much larger derivations to the ideal value at large I_{DL} . It can be attributed to the reduced voltage head room for REFS branches and increased voltage fluctuations in the RS node in option-B. As shown by Fig. 13(b), option-A outperforms option-B by five times reduced integral nonlinearity (INL) error and 3.75 times reduced differential nonlinearity (DNL) error.

V. MEASUREMENT RESULTS AND PERFORMANCE EVALUATION

Fig. 14 shows the photograph and the layout of the proposed macro. Its key specifications are summarized in Table II.

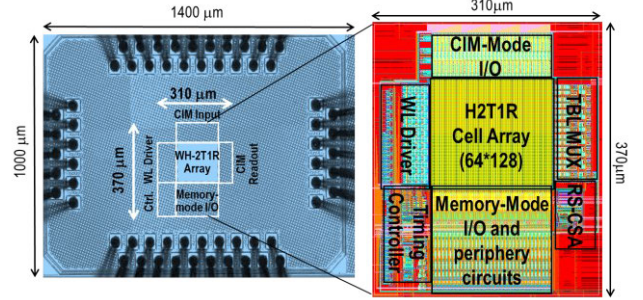


Fig. 14. Die photograph of the fabricated test-chip and macro layout.

TABLE II
TEST-CHIP SUMMARY

	Specification
Technology	28nm HKMG Logic Process
RRAM	Foundry TaOx-based RRAM
Bit-Cell	WH-2T1R (T1: I/O transistor, T2: core transistor)
Macro Size	310 × 370 μm^2
Array Size	64 rows × 128 columns (8Kb)
Mode	Memory/CIM
Power Supply (VDD)	0.8V (Core), 1.2V (IO)
Precision	1bIN-3bW-4bO
MAC EF	30.34 (TOPS/W)*1
Estimated MAC EF	154.04 (TOPS/W)*2

*1. Measured EF at VDD = 0.8V with 87.5% input sparsity

*2. Estimated EF at VDD = 0.8V with 87.5% input sparsity based on optimal T_{AC}

The test-chip is fabricated using the 28-nm HKMG logic process with back-end-of-line (BEOL) integrated TaOx RRAM devices. The macro integrates 8-kb 2T1R cells, and the area of the entire macro is 310 × 370 μm^2 . The power supplies (VDD) for the core and IO devices are 0.8 and 1.2 V, respectively. It can perform analog MAC operations with 1-bit-IN, 3-bit-W, and 4-bit-O in each clock cycle. The measured MAC EF is 30.34 TOPS/W, and the estimated one after readout timing optimization is 154.04 TOPS/W, which will be discussed later in this section.

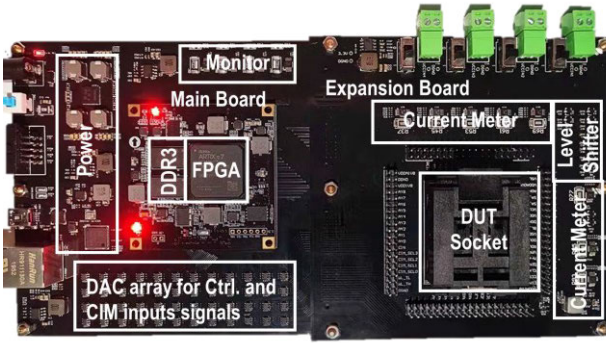
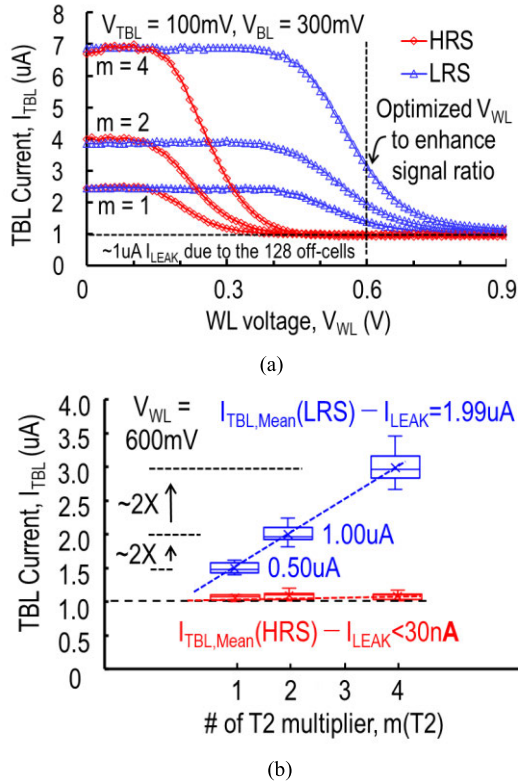


Fig. 15. Customized evaluation board for the test-chip.

Fig. 16. Measured bit-cell characteristics from the different SUBAs of the 64×128 cell array. (a) I - V curves. (b) I_{TBL} distributions.

A customized evaluation board is developed, as shown in Fig. 15. A field-programmable gate array (FPGA) as the host and an array of DACs as the analog inputs are integrated on the main board. The test-chip with the quad flat package (QFP) is equipped in the socket for testing on the extension board.

Fig. 16 shows measured bit-cell properties in the different SUBAs of the cell array. As shown in Fig. 16(a), I_{TBL} decreases as V_{WL} increases because of the voltage change at V_X . A clear memory window between the HRS and LRS cells can be observed at $V_{WL} = 0.6$ V. Note that there is I_{LEAK} about 1 μ A, which is due to the off-cell leakages of the 128 cells in the same row, as previously discussed in Section III. I_{LEAK} can be compensated using the leakage compensation branch in RS-CSA. Fig. 16(b) shows the measured I_{TBL} distribution at the optimized read conditions. After subtracting I_{LEAK} , the

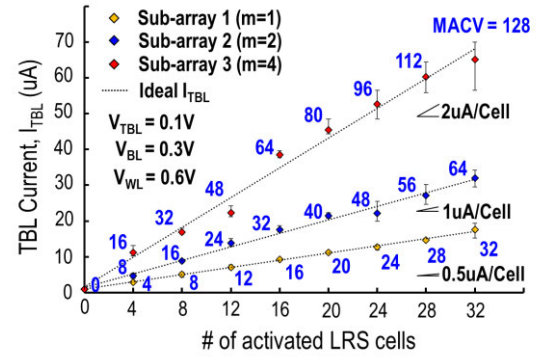


Fig. 17. Measured MAC current distribution in different SUBAs.

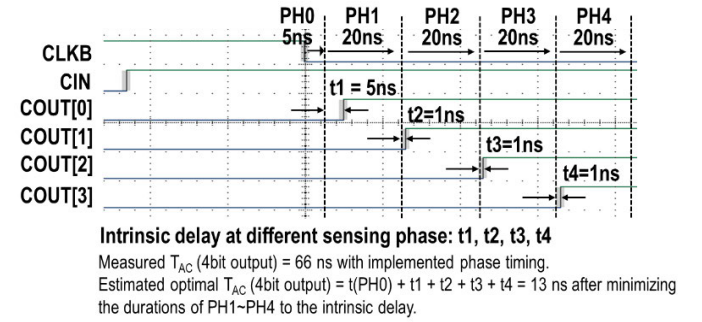


Fig. 18. Measured digital waveform of the macro.

mean values of I_{TBL} of the cells in SUBA1, SUBA2, and SUBA3 are 0.5, 1, and 2 μ A, respectively, which are well proportional to the number of multipliers of T2 in different SUBAs; I_{TBL} at HRS in different SUBAs is negligible below 30 nA, which can effectively suppress the MAC signal margin degradations due to the data-pattern variations.

Fig. 17 shows I_{TBL} distributions as a function of the number of activated LRS cells in SUBA1-3, which can be well fit by the ideal values based on the target cell currents. This result confirms the linear relationship between I_{TBL} currents and MACVs. The suppressed MAC signal non-linearity can be attributed to two results. First, the reduced cell currents of the 2T1R cells lower the summation currents. Second, the CIM data path has reduced parasitic resistance compared with that in the memory one due to that fact that no I/O devices are involved.

Fig. 18 shows measured digital waveforms of the macro from PH0 to PH4 in a typical computing cycle. Here, CLKB represents the opposite of CLK. After the falling edge of CLKB, the macro sequentially outputs the 4-bit data from COUT[0] to COUT [3] with a access time (T_{AC}) of 66 ns in total. Note that a long T_{AC} is because of the relaxed timing scheme in the current implementation for testing purpose, in which the durations of PH0 and PH1-4 are 5 and 20 ns, respectively. According to the measured waveforms, the minimized durations for PH1 and PH2-4 are 5 and 1 ns, respectively. Therefore, it is estimated that the minimized T_{AC} after timing optimization can be lowered to 13 ns.

TABLE III
COMPARISON WITH PREVIOUS RRAM CIM MACROS

	ISSCC 2021[28]	NE 2021[31]	JSSC 2022[33]	ISSCC 2022[35]	VLSI 2022[39]	JSSC 2023[37]	This Work
Cell Type	1T1R (RRAM)	1T1R (RRAM)	1T1R (RRAM)	1T1R (RRAM)	1T1R (RRAM)	1T1R (RRAM)	WH-2T1R (RRAM)
Technology	22nm	22nm	40nm	40nm	40nm	22nm	28nm
Computing Method	Current domain	Current domain	Voltage domain	Voltage domain	Current domain	Time domain	Current domain
Array Dimensions	1024*512 (*8)	512*512 (*8)	256*256	256*256	256*256	1024*256 (*32)	64*128
Summation Current Suppression	Current mirror downscale	Current mirror downscale	N.A.	N.A.	Current mirror downscale	Voltage-to-time converter	Current modulation by T2
HRS Leakage Suppression	N.A.	HRS Sensor	N.A.	Bias current compensation	N.A.	N.A.	13.5X enhanced R-ratio with sub-30nA HRS leakage
Read Disturb Suppression	N.A.	N.A.	On-line Disturb Detection	N.A.	N.A.	N.A.	-67% WL Voltage, -17% voltage drop on RRAM cells
Multi-bit Weight Processing	Digital Shifter & Adder	BL Voltage & Current Mirror Ratio	Digital Shifter & Adder	N.A.	CM-based subtractor	Digital Shifter & Adder	Number of the T2 multiplier, (m=1, 2, 4 for 3-bit encoding)
Analog-to-Digital (A/D) Readout	2-bit Voltage SA ($N_{REF}=2$)	2-bit Current SA ($N_{REF}=2$)	4-bit Flash ADC ($N_{REF}=15$)	4-bit Flash ADC ($N_{REF}=15$)	ADC-free analog computing	3-bit TDC ($N_{REF}=15$)	4-bit Current SA ($N_{REF}=4$)
# of Analog Accumulations per I/O	4	1	9	16	8	8	32
Classification Accuracy	N.A.	N.A.	96% (MNIST) 54% (ImageNet)	N.A.	89% (Cifar10) 67% (Cifar100)	91.74% (Cifar10) 67.11% (Cifar100)	89% (Cifar10)
MAC EF (TOPS/W)	195.7 @1bIN-2bW-4bO	36.61 @4bIN-4bW-11bO	56.67 @1bIN-1bW-4bO	26.56 @1bIN-1bW-4bO	26.97 @AnoIN-2bW-AnalogO	21.6 @8bIN-8bW-19bO	30.34 ^{*1} -154.04 ^{*2} @1bIN-3bW-4bO
1bit Normalized EF (TOPS/W*bit ²)	391.4	292.42	56.67	26.56	431.52	416.5	91.02 ^{*1} -462.12 ^{*2}

*1. Measured EF at VDD = 0.8V with 87.5% input sparsity *2. Estimated EF at VDD = 0.8V with 87.5% input sparsity based on optimal T_{AC}

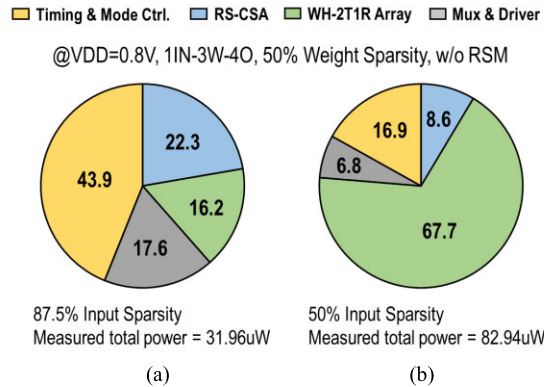


Fig. 19. Measured macro power and its breakdown. (a) 87.5% input sparsity case. (b) 50% input sparsity case.

The power of the macro varies depending on the sparsity in the data pattern. Fig. 19 shows the power and its breakdown of the macro in the cases with different input sparsity, which is measured based on the split power pins implemented in the test-chip. In the case of 87.5% input sparsity [Fig. 19(a)], the measured total power is 31.96 μ W. Because of the reduced power of the proposed cell array and RS-CSA, the timing and mode control accounts for the majority (43.9%) of the total power. On the other hand, in the case of 50% input sparsity [Fig. 19(b)], the measured total power increases to 82.94 μ W. It can be mainly attributed to an increased proportion (67.7%) of power consumed by the MAC array at high parallelism. In particular, the LRS cell currents on the activated BLs lead to the increased power of the MAC array.

In Fig. 20, we evaluate the EF of the proposed macro for deep neural network (DNN) processing using a typical ResNet-18 model trained on the CIFAR-10 dataset. The EF

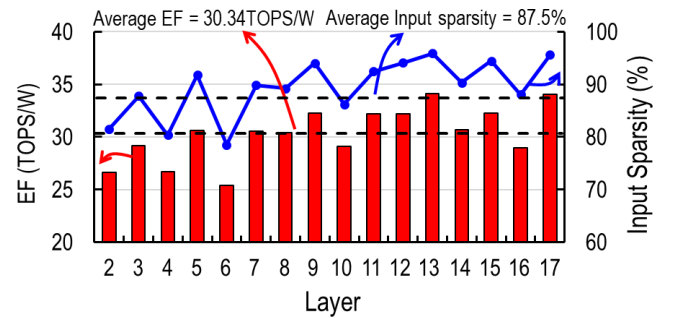


Fig. 20. EF of the proposed RRAM nvCIM macro across different layers with different input sparsity in the ResNet-18 model trained on the CIFAR-10 dataset.

varies layer-by-layer mainly determined by the input sparsity. The average input sparsity over different layers is 87.5%, which is corresponding to the case shown in Fig. 19(a). The average EF of the proposed macro for ResNet-18 processing is 30.34 TOPS/W. Besides, it can be further improved to 154.04 TOPS/W considering the optimized T_{AC} after timing optimization.

To evaluate the influences of the RSM scheme on the typical DNN tasks, we have performed simulations on the inference accuracy using an open-source benchmark framework for the CIM accelerators [46], [47] with the modified quantized method, input parallelism, input/weight/output precisions, and device parameters based on the conditions of our test-chip. The quantized model has 3-bit signed weights. To process the signed weights, all the weight data are changed to the unsigned data by adding 4. In a typical computing cycle, after the MAC operations in the proposed macro are finished, the MAC results are subtracted by four times of the number of activated inputs

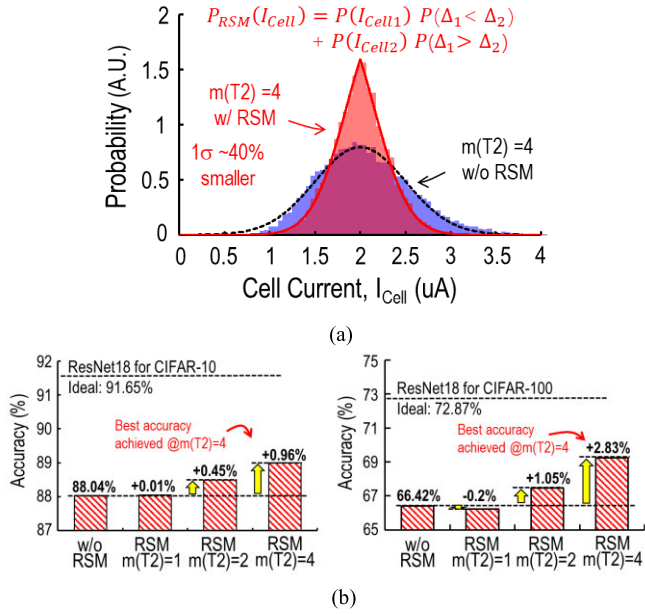


Fig. 21. Performance of the proposed RSM scheme. (a) Cell current distribution w/ and w/o RSM. (b) CIFAR-10 and CIFAR-100 inference accuracy by applying the RSM scheme on different SUBAs.

to obtain the MAC results with the signed weights. Moreover, when the RSM scheme is applied to a given sub-array, the corresponding device variation is set to be $P_{RSM}(I_{Cell})$, and the rest of them are set to be $P(I_{Cell})$. Fig. 21 shows the impacts of different implementation methods of the RSM scheme. Fig. 21(a) shows the probability distribution of the cell current of the cell with $m(T2) = 4$ before [$P(I_{Cell})$] and after applying the RSM scheme [$P_{RSM}(I_{Cell})$]. A 40% reduction on 1σ is achieved by the RSM scheme indicating its impact on removing the tail bits. Fig. 21(b) further shows comparison of the results of applying the RSM scheme to different SUBAs, which represents different place values in the multi-bit weight. Among different SUBAs, applying the RSM scheme to the MSB cells (MSB-RSM) as our current implementation shows the best result, which improves the inference accuracy by 0.96% and 2.83% on the CIFAR-10 and CIFAR-100 datasets, respectively. The result also indicates that the MSB-RSM scheme can play a more important role as the inference tasks become more complex. Besides, compared with applying the RSM scheme to all sub-arrays, the MSB-RSM scheme can reduce the overhead on area and power of the macro. The area and power overhead caused by the MSB-RSM scheme, including the additional sub-array, its periphery multiplexers (MUXs), and drivers, are 16.98% and 9.08%, respectively. Note that its power overhead is relatively small because the RSM scheme does not change the total number of activated LRS cells.

Table III shows comparison of the proposed 28-nm RRAM nvCIM with the recent silicon verified ones. This work has first demonstrated RRAM nvCIM macro based on the WH-2T1R cell array. Compared with previous works, it solves the challenges caused by the large cell current, small R -ratio, HRS leakage, read disturbance, and multi-bit processing at the cell level. Besides, compared with the previous analog

readout circuitry, the proposed scheme shows reduced cost for reference generation. On this basis, this work achieves a large number of analog accumulations among the presented works. Finally, the maximum EF of the proposed macro is among the highest level compared with the previous works.

VI. CONCLUSION

This work reports a 28-nm RRAM nvCIM macro design to boost the input parallelism, computing precision, and EF of analog MAC operations for intelligent edge devices. For the cell array, the proposed WH-2T1R structure outperforms the conventional 1T1R cell array in several aspects. First, it enables a decoupled CIM data path built with logic core devices with reduced parasitic effects. Second, it can achieve 88% reduced LRS current, 13.5 times enhanced R -ratio, suppressed read disturbance, and more than 63.4% reduced MAC power. Finally, it supports 3-bit weight encoding in the analog domain, in which the RSM-MSB is deployed to reduce devices' variations. As for the readout circuit, the proposed RS-CSA can achieve more than 79.5% power reduction due to the reduced number of stand-by reference signals and halve the CM working range to extend its dynamic linear range. The test-chip performs highly linear analog MAC operations over 32 accumulation channels with 1-bit-IN, 3-bit-W, and 4-bit-O. It achieves an average EF ranging from 30.34 to 154.04 TOPS/W. Compared with previous silicon-verified ones, this work has first presented a WH-2T1R RRAM nvCIM chip, which sheds light on further improving the parallelism, precision, and efficiency for analog MAC operations.

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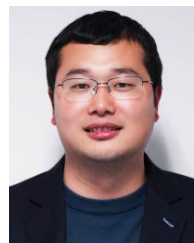


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